Joaquin Royer

Systems Engineer

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joaq.dev

Skills

- System Architecture
- AI, UI, Graphics Programming
- Networking/Replication
- Tool Development

Tools

- Unreal Engine 5, Unity
- C++, C#, Shader Graph, Blueprints
- Version Control
- WPF, XAML

Professional Experience

Limbitless Solutions, Orlando Fl — Aug 2023 - May 2026

- Non-profit that creates personalized EMG prosthetics for children
- Lead three and contributed to ten projects, three of which were Alt-Ctrl-GDC Finalists
- Worked with a custom hardware team and a medical research team to conduct research and create effective training using an electromyography game controller

Redline – Lead Programmer, *Alt-Ctrl-GDC Finalist* — Aug 2023 - May 2026

- Programmed a multiplayer racing mobile game in Unity for children with limb difference
- Created and documented a plugin allowing for integration of a custom EMG device into Unity's Input System, reducing development time by 20+ hours per project
- Leveraged Mobile GPU profiling tools to improve performance by 172%
- Prototyped reinforcement learning enemy AI using Unity ML Agents

Bionic Bash – Programmer, *Epic MegaGrant* — Jan 2025 - May 2025

- Authored editor tools that allowed custom controller connection without packaging, saving ~15 minutes per iteration
- Made a multi-platform Bluetooth LE plugin in C++ allowing connection to custom EMG devices on both PC and Mobile
- Implemented a local multiplayer UI system, allowing 4 players to have individual UI stacks
- Designed a round-based turn system that allows for interruptions and gameplay events
- Developed a minigame with students from the University of Skövde during a rapid prototyping game jam

Quantum's Pursuit – Programmer, *Unity for Humanities Grant* — Aug 2024 - Dec 2024

- Architected hierarchical state machine logic for 3 multi-stage boss fights
- Created an extendable power system allowing game powers to easily bind to input
- Refactored code to reduce six years of technical debt to allow for multi-platform support

Flex Desktop – Lead Programmer — June 2025 - Sept 2025

- Developed a WPF windows application in C# and XAML that allows mapping of custom game controller to a virtual X-Input device
- Implemented the Windows Game Input API in C++ to increase device support
- Architected the software using MVVM design patterns to ensure decoupled UI

Unboxing – Lead Programmer — Oct 2025 – Mar 2026

- Engineered an embedded ESP 32 C++ program that allows users to control RC vehicles with custom EMG game controllers
- Ensured memory safety in a performance critical environment
- Designed and built a touch UI experience using LVGL

Limbitless Journey – Programmer, *Alt-Ctrl-GDC Finalist* — Nov 2024 – Dec 2024

- Ported from Unity HDRP to URP to allow for Meta Quest Support
- Implemented mixed reality game object spawning and pathfinding, allowing software level generation and character movement to respect real-world objects

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Personal Projects

Sentinel Games – *SORN* – Lead Programmer Aug 2025 - May 2026

- Optimized network replication, maintaining a stable bandwidth below 50KB/s per client
- Developed a mission system with a node-based editor, allowing designers to script logic without programmer intervention
- Designed editor tools allowing missions to start on any objective and be fast forwarded to speed up development times
- Created enemy AI systems using Unreal State Trees
- Wrote technical design documents to standardize architecture for the engineering team
- Built a customizable character system that allows for the syncing of one-shot animations across multiple skeletal meshes
- Designed a mecha character that follows Unreal pawn-controller frameworks, allowing it to be reused across players and ai controllers.
- Implemented player and AI abilities and attributes using the Gameplay Ability System
- Engineered an Event Bus system allowing for centralized networked and not-networked events that can carry a custom payload
- Programmed a tagged actor system, allowing for constant time actor lookups utilizing Gameplay Tags

Cloud Weaver – *Pursuit of Knowledge* – Lead Programmer Mar 2025 - May 2026

- Architected a steam system allowing for the central resource of the game to be transferred in-between actors following composition patterns, enabling rapid prototyping for designers
- Implemented a level streaming system to allow for seamless world transitions
- Profiled game performance, fixing key CPU bottlenecks and eliminating 70ms frame-time spikes
- Programmed dynamic shaders influenced by gameplay code using Shader Graph

Education

University of Central Florida Aug 2022 - May 2026

B.A. Digital Media – Game Design

Community Involvement

Game Dev Knights – President Jun 2023 - May 2025

- Hosted 8 game jams with an average of 25 participants
- Presented 5 workshops about game programming patterns and optimization